Tanudeep Ganguly

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Summary

Edge AI and Computer Vision Engineer with hands-on experience in designing, training, and deploying vision-based machine learning models on resource-constrained systems. Strong background in Python, PyTorch, and ROS, with practical exposure to sensor fusion, real-time perception, and low-light image enhancement for autonomous and robotic applications. Passionate about building efficient, reliable, and scalable perception systems for real-world edge deployment.

Skills

Languages: Python, SQL

Frameworks: PyTorch, ROS/ROS2

Modules: Opencv, Numpy, Pandas, Matplotlib, Seaborn, Scipy, Sci-kit learn

Tools and Environments: Linux, VS Code, Git, Docker, ROS/ROS2 Navigation Stack, TensorRT/ONNX, CUDA

Work Experience

TATA Electronics Pvt. Ltd. May 2024 - Present

AI Engineer

- Lead development and integration of large-scale computer vision models for industrial edge applications.
- Deploy and optimize models on production systems, ensuring performance and reliability under real-world constraints.
- Collaborate with cross-functional teams to scale vision pipelines and automate edge AI workflows.

HyperHorizon Technologies Pvt. Ltd. Feb 2024 - May 2024

Robotics Software Developer Intern

CUBO-BOTS Pvt. Ltd., Agartala, Tripura Jan 2024 - Jan 2025

Technical Specialist

Education

NIT Agartala Aug 2023 - May 2025

M.Tech in AI Specialization ***SGPA: 10/10 CGPA: 8.7/10***

Relevant Coursework: Artificial Intelligence, Machine Learning, Deep Learning, Advanced Data Structures and Algorithms (Python), Optimization, Robotics, Computational Linear Algebra, Information Retrieval, Natural Language Processing.

Published Research Works

- **Spatial-Attention Segmentation (SAS) for Secured AGV Navigation in Off-Road Environment[DOI:https://doi.org/10.1007/978-981-96-7614-9_16] :** Rapti Chaudhuri, **Tanudeep Ganguly**, Suman Deb
- **Unstructured Outdoor Terrain Segmentation Using SegNav for Safe AGV Navigation[DOI:10.1007/978-3-031-93688-3_25] : Tanudeep Ganguly**, Rapti Chaudhuri, Suman Deb
- **YOLO-D: A Domain Adaptive approach towards low light object detection[DOI:10.1016/j.procs.2025.04.562]** : Partha Pratim Das, **Tanudeep Ganguly**, Rapti Chaudhuri, Suman Deb
- **rA*: Re-Planned A* Technique for Point-to-Point Robot Navigation in Dynamic Environments[DOI:10.1109/iSES63344.2024.00077] : Tanudeep Ganguly**, Rapti Chaudhuri, Suman Deb
- **You Only Look Once in Dark: An Analytical Approach for Low Light Object Detection[DOI:10.1109/iSES63344.2024.00013] :** Sutapa Sen, Rapti Chaudhuri, **Tanudeep Ganguly**, Suman Deb
- **MLSTM: A Novel Modified Long Short-Term Memory Networks for Melody Synthesis Melody Generation[DOI:10.1007/978-981-97-6222-4_44] : Tanudeep Ganguly**, Sushmita Sharma, Suman Deb

Awards and Certificates

- **Achieved Best Paper Award at IEEE iSES 2024 and MICA November 2023**
- **Featured in Acknowledgment section of the book "That is ROS" ISBN 978 93 48106 60 5 for insightful contribution of implementation and integrating the custom build robots with ROS.**